UNIT 3 · TOPIC 3.5

Detecting Network Attacks

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following is an automated security tool whose primary function is to monitor network traffic and generate an alert when a known attack signature is matched?

- ○ A. Password manager
- ○ B. Network-based Intrusion Detection System (NIDS)
- ○ C. File compression utility
- ○ D. DHCP server

2. A network administrator wants to monitor all inbound traffic at a corporate gateway and automatically block packets that match known malicious patterns, not just alert on them. Which tool should the administrator deploy?

- ○ A. Intrusion Detection System (IDS)
- ○ B. Network packet analyzer (sniffer)
- ○ C. Intrusion Prevention System (IPS)
- ○ D. Network Address Translator (NAT)

3. Which of the following correctly identifies the key difference between signature-based and anomaly-based intrusion detection?

- ○ A. Signature-based detection can identify zero-day attacks; anomaly-based detection cannot.
- ○ B. Anomaly-based detection is always less accurate than signature-based detection.
- ○ C. Signature-based detection matches traffic against known attack patterns; anomaly-based detection flags deviations from an established behavioral baseline.
- ○ D. Signature-based detection works without any prior configuration; anomaly-based detection requires hardware support.

4. A financial services firm frequently faces zero-day exploits targeting its proprietary trading platform. The security team is choosing between deploying a signature-based IDS and an anomaly-based IDS. Which detection method should the team prioritize, and what is the primary justification?

- ○ A. Signature-based, because it produces fewer false positives and is easier to tune.
- ○ B. Anomaly-based, because it can identify deviations from a behavioral baseline without requiring a pre-existing signature for the attack.
- ○ C. Signature-based, because it is computationally cheaper and scales better to high-throughput trading traffic.
- ○ D. Anomaly-based, because it completely eliminates false positives once the baseline is established.

5. In Scenario 3B, the National Guard network technician must recommend detective controls for a satellite-linked public Wi-Fi network at a disaster shelter. The network must detect both external attacks and misuse by shelter occupants. Which combination of tools best addresses both threat surfaces?

- ○ A. A NIDS for real-time traffic monitoring combined with a SIEM for log aggregation and cross-device correlation.

- ○ B. Antivirus software installed on each shelter laptop only.
- ○ C. A perimeter firewall with logging disabled to reduce alert noise.
- ○ D. MAC address filtering to restrict which devices can join the network.

6. A security team's IDS generates 600 alerts per day, the vast majority of which are false positives caused by an internal vulnerability scanner running nightly. Which action best improves the usefulness of the IDS without creating detection blind spots?

- ○ A. Disable the IDS during the nightly scan window to eliminate the false positives.
- ○ B. Route the scanner's traffic around the IDS sensor to reduce the alert load.
- ○ C. Replace the IDS with a next-generation firewall, which generates fewer alerts.
- ○ D. Tune the IDS ruleset to whitelist the scanner's IP address and adjust sensitivity thresholds for its known traffic patterns.

7. A university deployed a signature-based NIDS campus-wide eighteen months ago. A post-incident review of three recent breaches reveals that in each case the NIDS generated zero alerts during the intrusion. Evaluate the most likely root cause and the architectural change that would most improve detection coverage going forward.

- ○ A. Signature-based detection cannot identify polymorphic or zero-day attacks; layering an anomaly-based detection capability alongside the existing NIDS would best address the coverage gap.
- ○ B. The NIDS sensor was placed incorrectly; relocating it to the DMZ will prevent future misses.
- ○ C. The NIDS failed due to insufficient hardware resources; replacing it with an appliance-grade IPS resolves the detection gap.
- ○ D. The NIDS functioned correctly; the breaches did not involve network-layer activity.

8. A security architect is evaluating two detection approaches for a large enterprise: (1) a NIDS in promiscuous mode on the core backbone, and (2) distributed HIDS agents installed on every endpoint. Which statement most accurately captures the trade-offs between these two approaches?

- ○ A. NIDS always has lower operational overhead than HIDS because it does not run on production hosts.
- ○ B. HIDS is strictly superior because it sees all traffic before and after decryption at the endpoint.
- ○ C. NIDS provides broad network-layer visibility but may be blind to encrypted lateral movement; HIDS can inspect host-level activity but introduces endpoint overhead and cannot observe cross-segment traffic without log correlation.
- ○ D. Once TLS inspection is enabled on the NIDS, HIDS agents become redundant.

9. A SIEM monitoring the disaster-shelter network (Scenario 3B) fires an alert at 03:47: a single internal host is generating outbound DNS queries at a rate of one per second, each resolving a unique high-entropy subdomain of the same parent domain. Evaluate what this pattern most likely indicates, and determine which detective control would have surfaced it earliest, before the SIEM correlation fired.

- ○ A. Normal nightly software update check; no detective control needed; close the alert.
- ○ B. DNS tunneling or C2 beaconing via subdomain randomization; a NIDS with DNS query-rate and entropy anomaly rules would surface it earliest, prior to SIEM correlation.
- ○ C. A misconfigured recursive DNS resolver producing spurious queries; syslog monitoring alone is sufficient.
- ○ D. A DNS-based DDoS amplification attack originating from inside the shelter; blocking all outbound UDP/53 permanently resolves the issue.

10. How does an AI-assisted threat detection system differ from a traditional signature-based IDS in its ability to identify unknown attacks?

- ○ A. AI operates faster than IDS because it uses hardware acceleration.

- ○ B. AI completely eliminates false positives by automating the analyst response workflow.
- ○ C. AI can detect novel attack patterns by learning from behavioral baselines without requiring predefined signatures for each threat.
- ○ D. AI is only useful after a human analyst has first identified and labeled the threat manually.

11. A network analyst reviews authentication logs and finds the following pattern: 847 failed SSH login attempts from a single external IP address over a four-minute window, followed by one successful login. Which attack type does this log pattern most directly indicate?

- ○ A. SQL injection
- ○ B. Man-in-the-middle attack
- ○ C. DNS cache poisoning
- ○ D. Brute-force credential attack

12. Which of the following log artifacts is the most direct indicator that a port scan has been conducted against a network host?

- ○ A. Large volume of outbound SMTP traffic from a single internal host.
- ○ B. Connection attempts to many sequential or random ports on a single target host from one source IP in a short time window.
- ○ C. High-rate DNS queries for unique subdomains of a single parent domain.
- ○ D. Repeated authentication failures on an internal file share from the same user account.

13. The National Guard technician (Scenario 3B) reviews the shelter network's NIDS alerts and finds: a flood of UDP packets targeting port 53 originating simultaneously from 23 different internal host IPs toward the same external destination. Applying detection techniques, which attack type should the technician classify this as, and what is the most appropriate immediate containment action?

- ○ A. ARP spoofing originating from inside the shelter; enable dynamic ARP inspection on all switch ports.
- ○ B. SSL stripping attack; enforce HSTS headers on the shelter's web application.
- ○ C. Brute-force SSH attack; block inbound TCP port 22 at the satellite link perimeter.
- ○ D. DNS amplification attack with internal hosts acting as unwilling participants or bots; restrict outbound UDP port 53 to only the authorized DNS resolver's IP.

14. A SIEM alerts on the following pattern for a single internal host: 18 outbound TCP connections to 18 different external IP addresses, all on port 443, each connection lasting under 3 seconds, occurring over a 90-second window. Without AI assistance, a human analyst must classify this behavior. Which classification is most defensible, and what additional NetFlow evidence would confirm it?

- ○ A. Normal web browsing; no additional evidence needed.
- ○ B. Data exfiltration via HTTPS; confirm by looking for large total outbound byte counts per session.
- ○ C. Potential C2 beaconing; confirm by looking for regular time intervals between connections and consistent small payload sizes per session in NetFlow records.
- ○ D. A port scan; confirm by looking for ICMP echo requests preceding each TCP connection.

15. An AI-powered anomaly detection system flags an internal developer's workstation as exhibiting highly anomalous network behavior. Investigation reveals the developer recently began running containerized microservice test environments that generate substantial inter-service traffic. How should the security analyst use the AI alert alongside manual investigation to correctly classify this event?

- ○ A. Correlate the AI alert with change-management records (was the container deployment authorized?) and container platform logs (does the traffic profile match expected container orchestration behavior?) before classifying the event as benign or malicious.
- ○ B. Dismiss the AI alert because container environments are well-known sources of anomaly false positives.
- ○ C. Accept the AI classification immediately and isolate the workstation from the network pending forensic review.
- ○ D. Escalate directly to the incident response team without further investigation because the AI flagged it with high confidence.

16. A threat hunting team analyzes log files from a recently compromised network and reconstructs the following sequence: (1) an internal workstation issued a DNS query for a domain registered 48 hours prior; (2) a TCP/443 connection to that domain was immediately established and maintained; (3) over the following 72 hours, the workstation sent 4.2 GB outbound in small encrypted bursts averaging 6 MB per transfer. Evaluate which attack lifecycle phase this evidence sequence represents, and identify all indicators of compromise (IoCs) present.

- ○ A. Initial access phase; IoCs present: the DNS query and the newly registered domain only.
- ○ B. Lateral movement phase; IoCs present: the internal DNS query and the total outbound traffic volume.
- ○ C. Exfiltration phase of an established multi-stage intrusion; IoCs present: domain age under 48 hours (newly registered domain infrastructure), persistent HTTPS C2 communication (beaconing pattern), and large cumulative outbound data transfer volume (4.2 GB).
- ○ D. Reconnaissance phase; IoCs present: the DNS query only.

17. An attacker conducts a slow-and-low data exfiltration campaign, transferring exactly 10 MB of stolen data per day over 90 days, deliberately staying below daily traffic anomaly thresholds. Evaluate the comparative effectiveness of AI-assisted anomaly detection versus daily human log review for detecting this specific attack pattern, and identify the primary reason for the difference.

- ○ A. Human analysts are more effective because they can apply organizational context that AI models lack when interpreting traffic data.
- ○ B. Both approaches are equally effective because modern SIEM dashboards surface all traffic anomalies automatically regardless of the detection backend.
- ○ C. Neither approach is effective; only network segmentation prevents slow-and-low exfiltration from reaching an external destination.
- ○ D. AI-assisted detection is more effective for this scenario because it can maintain continuous behavioral baselines over long time horizons and detect cumulative drift that human analysts reviewing daily snapshots are statistically unlikely to identify.

18. A SOC analyst receives the following log extract from the disaster-shelter network (Scenario 3B):

[03:47:12] SRC=192.168.4.87 DST=8.8.8.8 PROTO=UDP/53 QUERY=a1b2c3d4.maliciousdomain.org
[03:47:13] SRC=192.168.4.87 DST=8.8.8.8 PROTO=UDP/53 QUERY=e5f6g7h8.maliciousdomain.org
[03:47:14] SRC=192.168.4.87 DST=8.8.8.8 PROTO=UDP/53 QUERY=i9j0k1l2.maliciousdomain.org
[03:47:15] SRC=192.168.4.87 DST=8.8.8.8 PROTO=UDP/53 QUERY=m3n4o5p6.maliciousdomain.org

Evaluate what this log pattern represents, which detective control would most effectively surface it before SIEM correlation fires, and what limitation exists in relying solely on a standard perimeter firewall to contain it.

- ○ A. DNS tunneling or C2 beaconing via randomized-subdomain encoding; a NIDS with DNS query-rate and subdomain-entropy anomaly rules would surface it earliest; a standard firewall allowing outbound UDP/53 cannot inspect DNS payload entropy and would not block these queries without also breaking all legitimate DNS resolution for shelter users.
- ○ B. Normal recursive DNS lookups from a misbehaving resolver; a syslog monitor would surface it; firewalls fully contain misbehaving resolvers.
- ○ C. A DNS-based DDoS amplification attack; permanently blocking all outbound UDP/53 is fully sufficient with no limitations.
- ○ D. Normal behavior for a device running automatic security update checks; no detective control needed; close the alert.